

Biosafety 101

LaBella Lab
Wild Yeast Project



Image Credit: r/sciencememes, u/luigi77714

Quick Introductions

Introduce yourself!

*Name / Preferred Pronouns / Fun Fact**

**What song are you listening to on repeat? Do you have a pet? etc.*



Basic Biosafety - Outline

- Introduction
- Biosafety Levels
- PPE requirements
- Biological waste
- Emergency Response Procedures
- Standard Operating Procedures



What is Biosafety?

- WHO (World Health Organization): biosafety is the use of containment principles, technologies, and best practices to prevent unintentional exposure to pathogens and toxins, or their accidental release
- Biosafety is a PRACTICE: Biosafety practices are used in the proper handling of biohazardous organisms.
- Biohazardous organisms are microorganisms with an infectious potential for man, animals, and plants in the environment.
 - Includes prokaryotic and eukaryotic microbes, viruses, sub-viral infectious agents and recombinant organisms with any potential for survival in the environment or in living materials and can cause a health risk.



Biosafety Levels

- Levels define lab practices, equipment, and facilities needed to handle agents from low (BSL-1) to high (BSL-4) risk.
- Safety is cumulative
 - BSL-2 must follow all BSL-1, BSL-3 must follow BSL-2 and BSL-1

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Biosafety Level Chart



BSL - 1

- **Agents:** Non-pathogenic, well-characterized
- **Risk Level:** Minimal
- **Facility Requirements:** Standard laboratory

- **Practices & Safety Equipment:** Standard microbiological practices, PPE (gloves, lab coats)



BSL - 2

- **Agents:** Pathogenic
- **Risk Level:** Moderate
- **Facility Requirements:** Limited access, biohazard signs

- **Practices & Safety Equipment:** BSCs for aerosols, enhanced PPE, sharps precautions



BSL - 3

- **Agents:** Indigenous or exotic
- **Risk Level:** Serious
- **Facility Requirements:** Controlled access, negative air pressure

- **Practices & Safety Equipment:** BSCs for all activities, respirators, controlled access



BSL - 4

- **Agents:** Dangerous and exotic
- **Risk Level:** High
- **Facility Requirements:** Isolated facility, full-suit protection

- **Practices & Safety Equipment:** Full-body air-supplied suits, extensive decontamination

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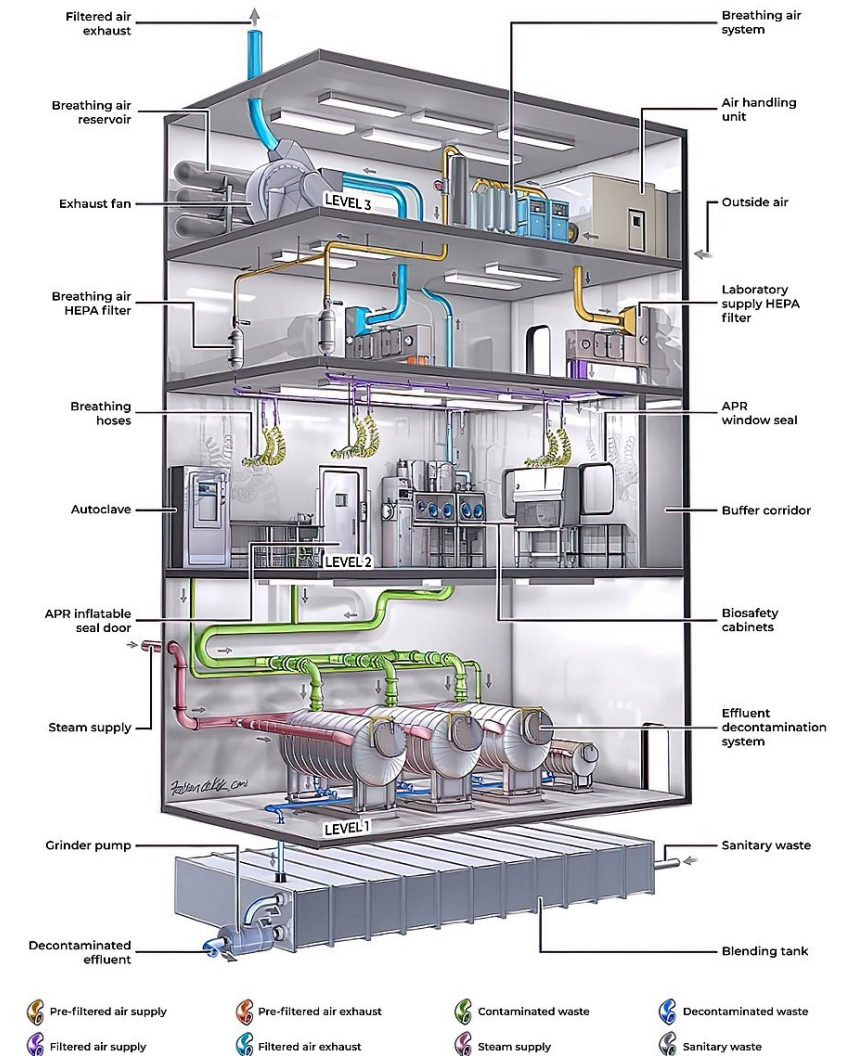
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BSL-4

- Designed to contain infectious agents that are easily transmitted (aerosol) and can cause fatal disease for which there are no known treatments or cures.
- Some NASA samples must go to a BSL4 facility
- Closest facilities to here are in Atlanta (CDC – one of two global facilities with smallpox) and Georgia State (B virus).

SCHEME OF THE MOST ISOLATED BIOLOGICAL LABORATORY
FOR WORKING WITH MICROORGANISMS
OF PATHOGENICITY GROUPS I-II



https://en.wikipedia.org/wiki/Biosafety_level



BSL-3

- Designed to contain pathogens transmitted via inhalation that are not readily treatable or preventable with vaccination
- Must contain a BSL-3 Biosafety Cabinet
- All personnel must have medical surveillance and offered immunizations.
- Room design such as unrestricted air flow, two sets of self-closing doors, no soft materials (carpets), and air must flow from clean to dirty air.



BSL-2

- Specific training in handling pathogenic agents
- Access to laboratory is restricted
- Procedures that create aerosols or splashes should be in a biosafety cabinet
- Extreme caution with sharps



BSL-1

- Work with organisms not known to cause disease in healthy humans
- Proper Personnel Protective Equipment (PPE)
 - Lab Coat (closed)
 - Close-toed shoes
 - Gloves
 - Eye protection
 - Hair tied back
- Use standard microbiological practices

BSL-1 lab

Work can be performed on an **open lab bench or table**



Lab coats, gloves, eye protection are worn as needed

BSL-2 lab

All procedures that can cause infection from aerosols or splashes are performed within a **biological safety cabinet (BSC)**



Lab coats and gloves are worn. Eye protection and face shields can also be worn as needed

BSL-3 lab

All work with microbes must be performed within an appropriate **BSC**



Laboratory coats, gowns, gloves, eye protection, face shields, shoe covers must be worn, and respirators might be required

BSL-4 lab

The laboratory is in a **separate building** or in an **isolated and restricted zone** of the building



Full body air-supplied positive pressure suit

BSL-2 Practices

The LaBella Lab is a **BSL-2 lab**, which means we work with potentially infectious agents. *If you are at increased risk of infection, we can adjust PPE or find alternate experiments.*

- Do not enter the lab unsupervised. Exceptions must be approved by Dr. LaBella or the relevant PI.
- No eating, drinking, smoking, cosmetic application, or phone use is allowed in the lab.
- Do not wear lab coats, gloves, or any other protective equipment outside of the lab.
- Use protective equipment (PPE) where required.
- Replace PPE that has torn or been spilled on.
- Follow all of the guidelines in the lab and institution biosafety, laboratory biosafety and environmental health and safety (EHS) protocols.



BSL-2 Practices

- Wash your hands after handling biologicals or hazardous materials, taking off gloves, and before leaving the lab.
- Disinfect all work surfaces daily and decontaminate any spills.
- Carefully handle all sharps and dispose of them properly.
- Minimize the formation of aerosols and splashes.
- Wear your PPE!!!
- Dispose of all biological and chemical waste properly.



! Repeated violations of lab safety practices will result in removal from the lab. !



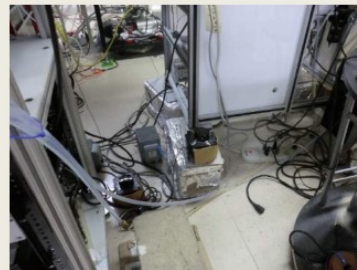
General Practice

- Pay attention to your surroundings!
- Keep your working area clean and tidy
 - “Lab Citizen”

- Egress pathways are the ways you exit the lab in an emergency. If pathways are blocked or obstructed, an employee would have a problem exiting the building.



Good



Bad



Tie back long hair, and wear suitable gloves, goggles, and other personal protective equipment. Avoid touching your eyes, nose, and mouth.

Proper supervision

Don't perform lab experiments without instructor supervision unless given permission to do so.



Know location of emergency numbers & safety equipment

Know the location of safety equipment and emergency phone numbers (such as poison control) so you can access them quickly if necessary.



No food

Don't eat or drink in the lab, and never taste chemicals.



ID hazards

Identify hazardous materials before beginning labs.



Be attentive

Be attentive while in the lab. Don't leave lit Bunsen burners unattended or leave an experiment in progress.

Be careful when handling hot glassware

Turn off all heating appliances when not in use. Keep flammable objects away from your work space.



Keep a clean work space

Don't obstruct work areas, floors, or exits. Keep coats, bags, and other personal items stored in designated areas away from the lab. Don't block sink drains with debris.



Handle glassware carefully

Properly dispose of anything that breaks. Report cuts, spills, and broken glass to your instructor immediately.



Clean up

After completing the lab, carefully clean your work space and the equipment, and wash your hands with soap and warm water for at least 20 seconds.

Lab Safety Rules

Science labs offer great opportunities for learning, teaching, and research. They also pose hazards that require proper safety precautions.



Stay safe when conducting your labs by maintaining social distancing.

Sources: Carolina Biological Supply Company. "Lab Safety Dos and Don'ts for Students." <https://www.carolina.com/teacher-resources/interactive/lab-safety-instructions/tr35303.tr>.

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Waste Management

- Designated disposal for different lab waste types
 - Biohazard solids
 - Biohazard liquids
 - Solids
 - Chemicals
 - Sharps



LAB WASTE



Emergency Equipment

- Emergency safety shower and eye wash station are available
- Tested yearly (shower) or weekly (eye wash)
- Used in case of exposure to chemicals or some biohazards



Emergency Equipment

- Spill kits are specific to different types of spills
- May include absorbents, PPE, brooms, chemical containers, and other neutralizers



Emergency Equipment Locations

- First Aid Kit– marked box, next to lab notebooks
- Emergency Eye Wash – stations at sink and next to -80C freezer
 - 15-minute wash with assistance
- Chemical Safety Shower – next to -80 C freezer
 - Must strip to undergarments and rinse for 15 minutes. Call 911 if necessary.
 - Safety before modesty!
- Spill Kit – next to chemical safety shower

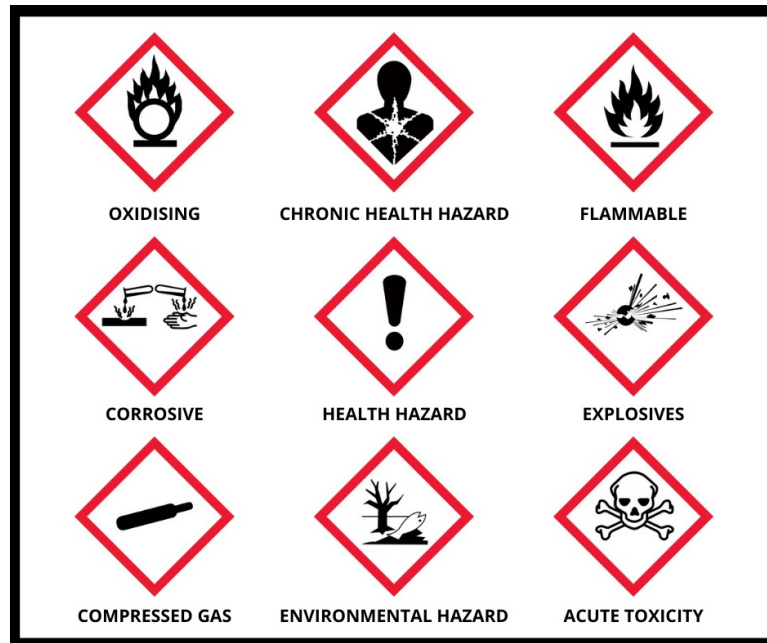


Image Credit: RedBubble; u/gracecotton99



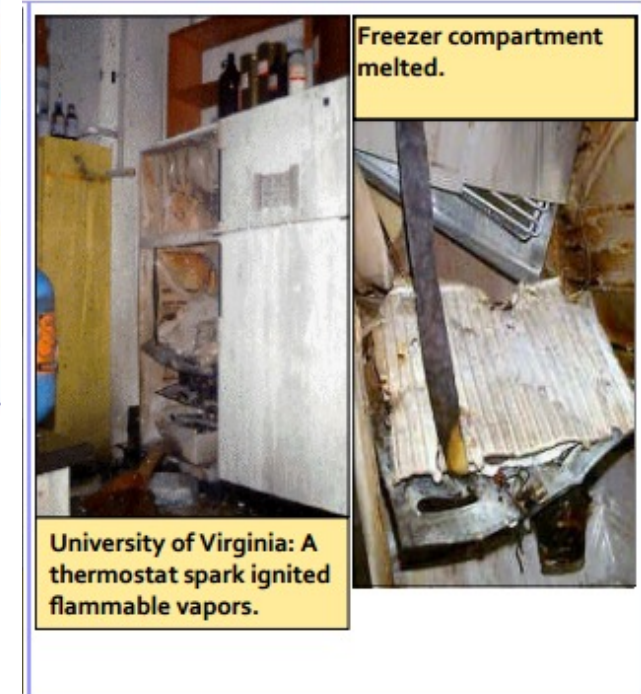
Chemical Safety

- Flammable, corrosive, toxic, or reactive chemicals are stored in approved, labeled cabinets.



Chemical Safety

- Flammable, corrosive, toxic, or reactive chemicals are stored in approved, labeled cabinets.
- **DO NOT PUT FLAMMABLE LIQUIDS INTO THE REFRIGERATOR**
 - Includes ethanol, isopropanol



Microbial Aseptic Technique

- Microorganisms are *everywhere* – and can contaminate our samples
- We want to reduce the likelihood of contamination with best practices!



Aseptic Technique

- Wipe down your working area with 70% ethanol **before and after** working
- Use only sterile reagents, tips and tubes
- Minimize contamination from clothing and body surface.
- Avoid touching or breathing on sterile surfaces.
- Avoid talking when transferring materials or opening containers
- **Work quickly and CLOSE LIDS AS SOON AS POSSIBLE**
- Other labs may work over a Bunsen burner – we do not have that capability so it is extra important that you follow the other guidelines



Before you act...

- Do you have all tubes and tips ready to go?
- Do you have enough hands to hold everything you need?

Think it through!



Lab Safety Training

- **Laboratory Personnel BSS Training**

<https://research.charlotte.edu/departments/office-research-protections-and-integrity-orpi/biosafety/biosafety-training-information/>

- **Lab Biosafety Protocol**

https://drive.google.com/file/d/1kZKfBEeVSiTNbXMYxc7CZZ0bgad2wRc7/view?usp=drive_link

- **Lab EHS Manual**

https://drive.google.com/file/d/17Q_9g7UkfkQCxjsH2HckuvDxuQMfjhlr/view?usp=drive_link

- **Lab Basics**

https://github.com/The-Lab-LaBella/Wild_Yeast_Summer_2026/blob/main/lab_basics.md

